

Evidence-Based Generative Design

Tool Pathways for Studio Decisions

Generate comparable options, not renders — with a workflow you keep.

Hongshan Guo · 12 weeks × 2 hours · undergraduate, MArch, and PhD in one room



One question runs the whole semester

Final review. “The atrium improves daylight.” A critic asks: *by how much? compared to what?*

What evidence would make this design decision more defensible?

Where studio decisions usually come from — all of us, no exceptions:

- precedent and intuition · tutor pressure, or taste
- assumptions nobody tested · “evidence” added after the decision, to justify it

The reframe: not “*what software should I use?*” but “*what evidence would reduce uncertainty around **this** decision?*” — and then the shortest workflow that produces it.

Straight talk: this is not last year's course

Last year's feedback was clear — too philosophical, not useful enough. We rebuilt the semester around the tools.

Was

course philosophy first
a “research brief” to write
theory readings
two loose tracks (no/low-code vs code-heavy)

a machine-learning theory detour
first working build due wk 5

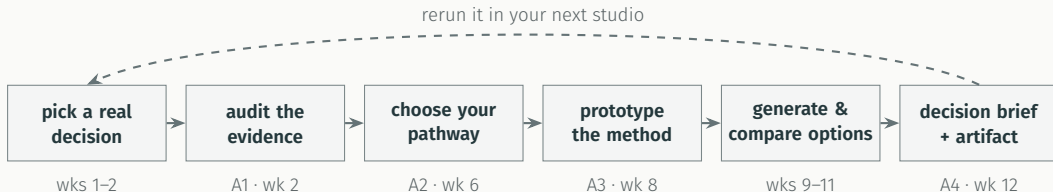
Is now

tool landscape first — week 1 is a working session
a studio decision you actually face
named drills, every single week
**one assessed pathway you choose: Spreadsheet, GIS,
or Python**

Python = notebooks you adapt; ML strictly optional
first working build due wk 8 — real time to build

What we kept: the four-assignment arc and its weights, mentor consultations, and the week-12
Object Fair.

How you will actually generate



Generating means producing *comparable options*, not renders:

- **Spreadsheet** — score options A/B/C on daylight, access, cost, risk → a decision matrix
- **GIS** — three candidate sites × 400 m buffers and walksheds → a defensible site pick
- **Python** — 200 design variants → cooling, daylight, and glare trade-off charts (dataset provided, or your own variant sweep — you write the analysis)

“Generative” here is a workflow you can rerun — evidence in, defensible options out.

Choose one pathway — go deep, keep the others legible

SPREADSHEET

Excel · Google Sheets

Best when the decision needs **comparison, ranking, visible logic**.

You build: evidence log → alternatives → charts with ranges → decision matrix.

The baseline exit pathway. Not remedial — the lowest-friction route to a decision-ready workflow.

GIS

QGIS · ArcGIS · geopandas

Best when **site, access, zoning, or overlap** drives the decision.

You build: buffers, walksheds, spatial joins → one clean map per question.

Week 4 compares all three head-to-head — and these are the outputs feasibility and master-planning teams actually use.

PYTHON

Jupyter · pandas · Colab

Best when **repeatability, reuse, or scale** matters.

You build: notebook analyses adapted from the course starters — never *required* to start from a blank file.

“If a spreadsheet answers the question cleanly, that is still a strong choice.”

The week-6 commitment test — if you cannot finish this sentence, you have not chosen yet: *“I am choosing ____ because my decision depends primarily on ____, and the minimum useful output will be ____.”*

Pathway choice should feel boring: the shortest path to a useful output — never the most impressive tool.

BIM, BEM, performance: literacy for everyone, pathway for no one

- **BIM** answers *what do we build and how does it fit together*; **BEM** answers *how will it behave over time*
- translation between them loses geometry, zones, constructions, operations — **translation loss is normal, not an edge case**
- simulation is evidence — useful, never automatically authoritative
- the paired-run rule: simulate your variants A and B — the *delta* outlives translation error

Week 5 drill — Translation Detective. A five-storey office exports to gbXML (the standard BIM→energy-model handoff) and reports **3× the expected energy use**. Your team finds the failure before anyone trusts the number. The week's real-scale case: **465 thermal zones** where ~30 belong.

The week's lesson: *"The file opened" did not mean "the model is valid."*

Taught to everyone — but your assessed pathway stays Spreadsheet, GIS, or Python.

Four buildings we keep interrogating

These four recur all semester as standing prompts: *what evidence would you want before defending this claim in public?*



The Edge
Amsterdam

wk 11 review lens:
performance claim — method,
or branding?



High Line
New York

wk 3: what would your
spreadsheet need to track?



**California Academy of
Sciences**
San Francisco

wk 5: what evidence backs
the envelope claim?

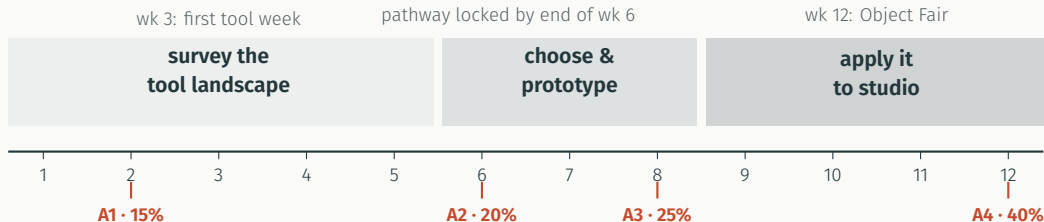


Kai Tak Sports Park
Hong Kong

wks 4 + 6: the local site,
access, and urban evidence
case

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Twelve weeks, three phases, four deadlines



- **weeks 1–5** — audit your own evidence, then survey spreadsheet, GIS, and BIM/BEM literacy with drills each week
- **weeks 6–8** — commit to one pathway, build and pilot the smallest workflow that answers a real piece of your question
- **weeks 9–12** — figures, thresholds, a structured draft review, and a public Object Fair with critique

You make things in class — every week

wk

- 1 sort your own past evidence — measured, observed, simulated, cited, tested by making, assumed — and leave with **A1 already underway**
- 2 turn one prior decision into a full evidence audit; A1 due
- 3 15-minute **Workbook Build Sprint** on your own topic, then the **Peer Swap Test** — can a classmate read your logic?
- 4 **Tool Choice Drill** (QGIS vs ArcGIS vs geopandas) + a live geopandas demo: load, inspect CRS, filter, buffer, plot, export
- 5 **Translation Detective** — diagnose a BIM→BEM export before trusting its numbers
- 6 **Pathway Commitment Drill** + in-class A2 work session; A2 due
- 7 **10-Minute Method Challenge** + **Documentation Swap** across pathways, no verbal explanation allowed
- 8 pilot your method; upgrade the language: *“the pilot suggests”*, never *“the data proves”*; A3 due
- 9 **Figure Rescue Sprint** + **Option B Drill** — the 3-second test; your variants count as evidence
- 10 **draft review** — pressure-test the near-final package, leave with a revision plan
- 11 **Evidence Package Workshop** — peer review against four questions before anything is final
- 12 **Object Fair** — public pin-up: your evidence board, your recommendation, your limits; A4 due

Four assignments, one thread: a decision you actually face

	due	weight	the question it answers
A1 Evidence Audit + Next Decision Brief	wk 2	15%	what did I really know — and what do I need next?
A2 Evidence and Pathway Map	wk 6	20%	which sources, which workflow, and why?
A3 Method Prototype	wk 8	25%	can I run a credible pilot in my pathway?
A4 Studio Evidence Package	wk 12	40%	can someone act on this?

- each assignment builds directly on the previous one — same project, rising stakes
- A3's standard: *"The purpose is not scale. The purpose is proof"* that your pathway can produce decision-relevant output — a working artifact that is **interpretable, not polished**
- no exams, no quizzes — the four assignments are 100% of the grade, $\approx$ 50–70 hours outside class all semester; drills are practice, not graded, but they build your assignment material

Where it lands: the Object Fair, week 12

A4 — the Studio Evidence Package (40%) has four parts that reinforce each other:

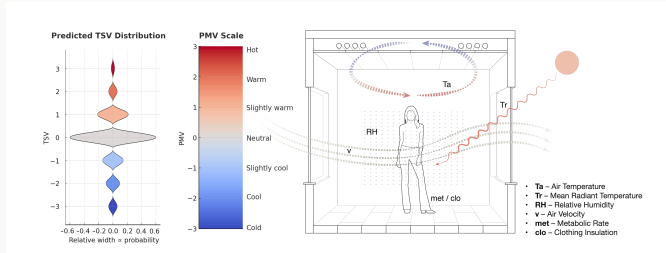
1. a **studio decision brief** (6–10 pages, written for critics and practice stakeholders)
2. an **evidence board / object card** — one poster-scale page that survives 30 seconds of a stranger's attention
3. the **pathway artifact** — your working workbook, map set, or notebook; it proves the package came from a workflow, not from slides
4. a **method appendix** (2–4 pages) — enough for someone else to rerun or extend the work

At the fair you present five things:

the decision • the pathway • the evidence • the recommendation • the limit

This is not just a report. It is a decision-support package — and it is yours to reuse.

The standard for evidence: ranges, not single numbers



One room, two claims of very different strength: PMV (the standard single-number comfort score) vs. the predicted *distribution* of what occupants actually feel. (Figure from the instructor's research.)

- every pathway ends the same way: **uncertainty shown, not hidden** — ranges on key figures are in the A4 rubric, and transparency about limits is 25% of A3
- scenario bands in the workbook · stated extents and assumptions on the map · confidence intervals in the notebook

You do not start from zero

In the repo on day one

- **starter notebooks** you adapt — you never have to start blank: an A/B pilot analysis with error bars and an object-card starter figure, plus an optional modeling extension
- **three practice datasets**, openly synthetic: 200 design variants (cooling / daylight / glare), a 120-row A/B pilot log, and 60 occupant survey comments — swap in your own data with the same structure
- the four-image precedent bank, licensing already sorted

Setup ladder

- *“use the lightest setup that still lets you complete your chosen pathway well”*
- spreadsheet access — everyone
- QGIS — only if you choose GIS
- Anaconda + Jupyter — only for Python or code-based GIS (geopandas); **Colab** is the no-install fallback

Local data that is actually there

DATA.GOV.HK, Planning & Transport Departments, HK Observatory climate records, OpenStreetMap

Humans: office hours (setup + pathway troubleshooting) · in-class work sessions · peer review

The takeaways are explicit — and they are tools

The syllabus states five learning outcomes; this is the form you actually hold them in by week 12:

1. a **working workflow artifact** in your pathway — workbook, map set, or notebook — built on your own project, reusable in the next one
2. an **evidence audit habit**: measured / observed / simulated / cited / tested by making / assumed — applied before the decision, not after
3. **judgment about tool fit** — when a spreadsheet beats a notebook, when a map beats both
4. a **language for uncertainty and thresholds** that critics and clients take seriously
5. a **documented method** someone else could rerun — your method appendix

The closing standard: strong work never sounds like “*I used a tool.*”

It sounds like “*I used this workflow to reduce uncertainty around a real design decision.*”

What is graded — and what deliberately is not

Graded, in every rubric

- method-question **fit**
- **honesty**: A1 grades specificity & honesty at 35%; A3 grades transparency about limits at 25%
- a workflow that can be **rerun or explained**
- **decision usefulness** — could a stakeholder act on it?

Not graded, ever

- tool virtuosity or code elegance
- how many software packages you touched
- choosing the most intimidating tool — “*overreach*” means a tool the question doesn’t need, never depth within one it does

The tools are takeaways, not the graded object. Support is scoped to your one pathway — starter files, a setup ladder, office hours, peer swaps; we don’t train you across all three tools. Tiers are floors, not ceilings: opting into deeper expectations is never a grade risk. This is a course in deciding with evidence, not software training.

What this course is not

Worth knowing before pathways lock in week 6 — this course is **not**:

- **software training** — no feature tours, no certification; the deepest *required* code is a five-line geopandas workflow, the Spreadsheet pathway needs none, and the optional notebooks go further for anyone who wants depth
- **an AI/ML course** — Python here means pandas and notebooks; machine learning is optional depth for those who want it
- **a theory seminar** — no readings-driven weeks; every concept lives inside a drill, a template, or a rubric
- **a data-science conversion** — you stay a designer: the tools serve design decisions, they don't replace them

And honest pacing, so nobody is surprised: weeks 1–2 audit *your own* past decisions (working sessions, not software demos); the first tool week is week 3; your first full build lands in weeks 7–8.

Same arc for everyone — depth scales

	what stays common	what scales
Undergraduate	same weeks, same four assignments	tighter scope, more scaffolding
MArch	same weeks, same four assignments	stakeholder fit, independence, practice-ready outputs
PhD	same weeks, same four assignments	deeper uncertainty treatment, stronger reproducibility

- the difference is **depth and independence, not separate coursework**
- you are never rewarded for out-teching the room — only for fitting the method to the question in front of you

Come to week 1 with three things

1. **one design decision you already made** — a façade proportion, an entrance location, a shading strategy — plus whatever counted as evidence at the time
2. **one upcoming studio decision** worth investigating — a live thesis decision is explicitly welcome; the course runs alongside it
3. **working spreadsheet access** — Excel or Google Sheets, the whole week-1 setup

You will leave the first session with A1 already underway.

By week 12 you should be able to say:

“This is the decision, this is the evidence,
this is the method — and this is how confident I am.”

Everything lives on the course site: schedule, assignment briefs, starter notebooks, and setup guides — questions any time at office hours.